Day 10 Report: Generating Dog Images with a GAN

Generative AI - DCGAN trained from scratch on Stanford Dogs (64 x 64), Colab GPU

1. Goal

Train a **GAN** (Generative Adversarial Network) that creates **new dog images** from random noise, and show how the images improve during training. Then compare it with a pre-trained diffusion model.

2. How a GAN works (simple)

Two networks compete. The **Generator** is a forger that makes fake dog pictures. The **Discriminator** is a detective that tries to tell real photos from fakes. Each one gets better because the other one gets better.

GAN training loop



3. Data and setup

Item	Value
Dataset	Stanford Dogs (Kaggle), 20,580 photos with bounding boxes
Preparation	Cropped each dog with its bounding box, resized to 64 x 64
Images used	10,000 (GPU mode)
Generator / Discriminator size	3.6 M / 2.8 M parameters
Training	25 epochs, 156 batches per epoch, about 18 s per epoch (7.9 min total)
Tricks	Label smoothing (real = 0.9), checkpoints saved to Google Drive

4. Results



Figure 1. The same 64 noise vectors at epoch 1, 15 and 25

- **Epoch 1:** only a checkerboard pattern, no dogs.
- **Epoch 15:** dog-like colours, fur texture and shapes on grass or indoor backgrounds.
- **Epoch 25:** clearer dog bodies and faces in many images, but still blurry and distorted.

Epoch	D loss	G loss	Meaning
1	0.961	7.791	Detective wins easily, forger is bad
5	0.826	3.379	Forger catches up fast

Epoch	D loss	G loss	Meaning
15	0.815	3.217	Balanced
25	0.815	3.224	Still balanced, no collapse

The losses stayed **balanced** after the first epochs (D around 0.8, G around 3). This means training was stable: neither network completely beat the other, and the generator did not collapse to one image.

5. Diffusion model comparison

The Stable Diffusion part **did not finish** in this run: the cell stopped with *NameError: name 'time' is not defined*, because the runtime restarted and the import cell was not run again. Fix: run the first cell again (or Run all), then run the diffusion cell. Expected result: diffusion images are much sharper and higher resolution (512 x 512) but take several seconds per image, while the GAN makes 64 small images almost instantly.

6. Problems and fixes

- The first dataset was 81 GB and filled the Colab disk. Switched to Stanford Dogs (about 750 MB) and deleted the raw download after preparing the images.
- Limited free GPU time: images are cached as a .npy file and checkpoints are saved to Drive, so training can continue after a disconnect.

7. Limitations and next steps

- 64 x 64 is small and 25 epochs is short; more epochs would give clearer dogs.
- No number-based quality score yet (for example FID); judged by eye.
- Next: re-run the diffusion section and compare speed and quality side by side.